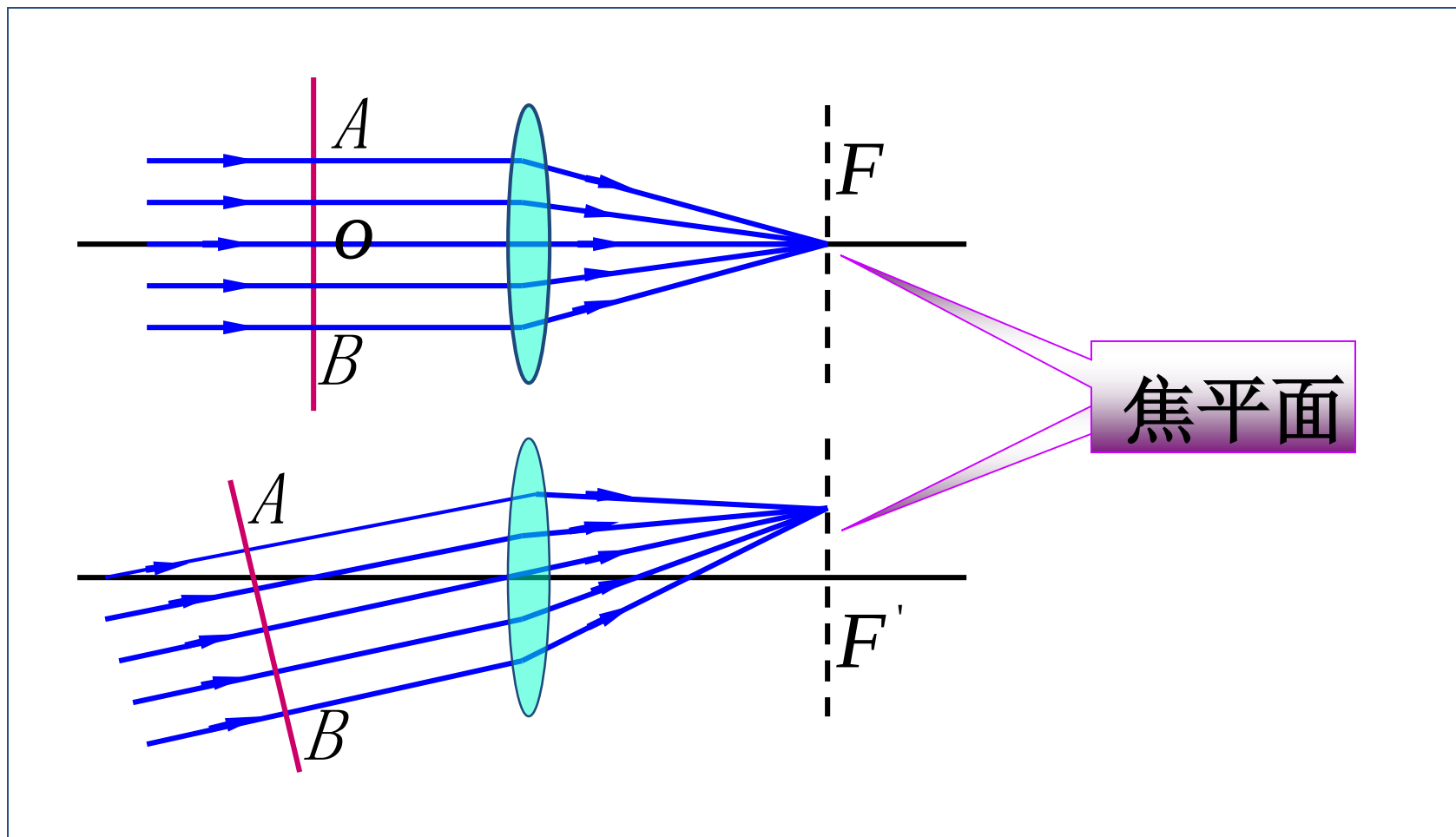




一 透镜不引起附加的光程差



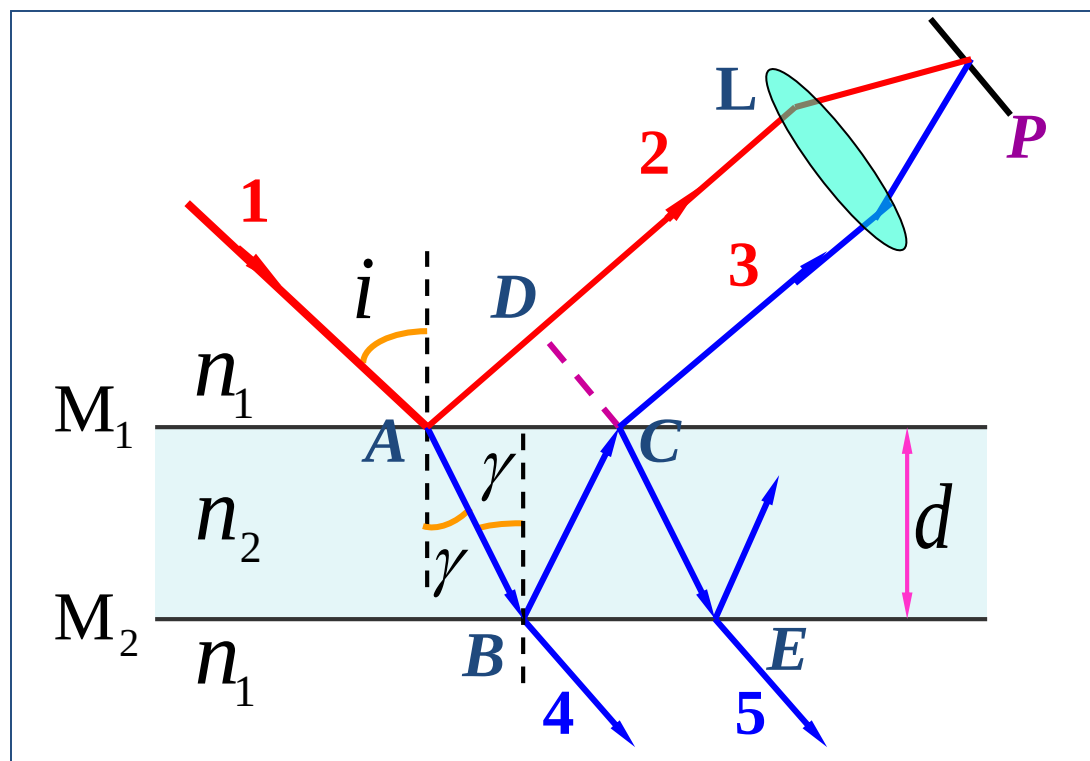


二 薄膜干涉的光程差

$$n_2 > n_1$$

$$CD \perp AD$$

$$\frac{\sin i}{\sin \gamma} = \frac{n_2}{n_1}$$

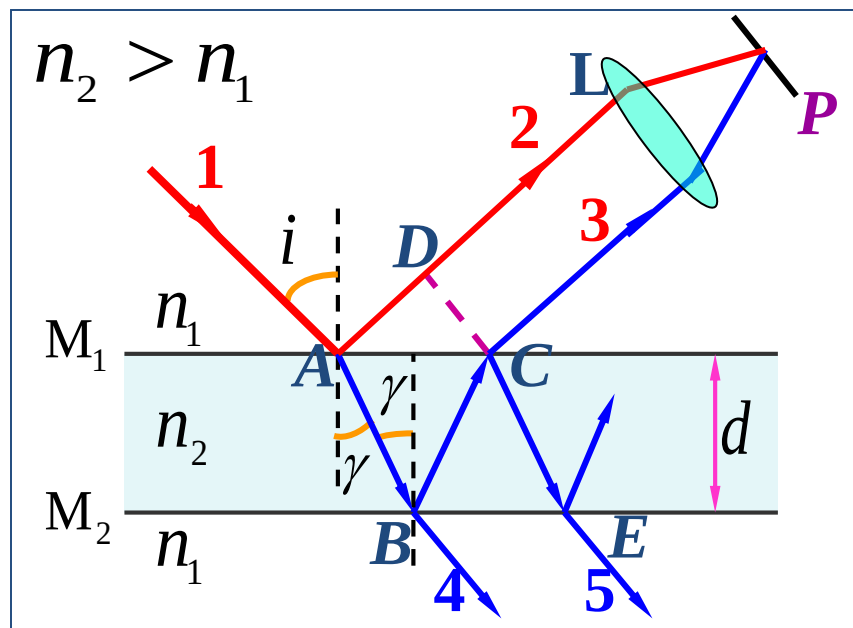




$$\Delta_{32} = n_2(AB + BC) - n_1 AD \quad \boxed{+ \frac{\lambda}{2}}$$

$$AB = BC = d / \cos \gamma$$

$$\begin{aligned} AD &= AC \sin i \\ &= 2d \cdot \tan \gamma \cdot \sin i \end{aligned}$$

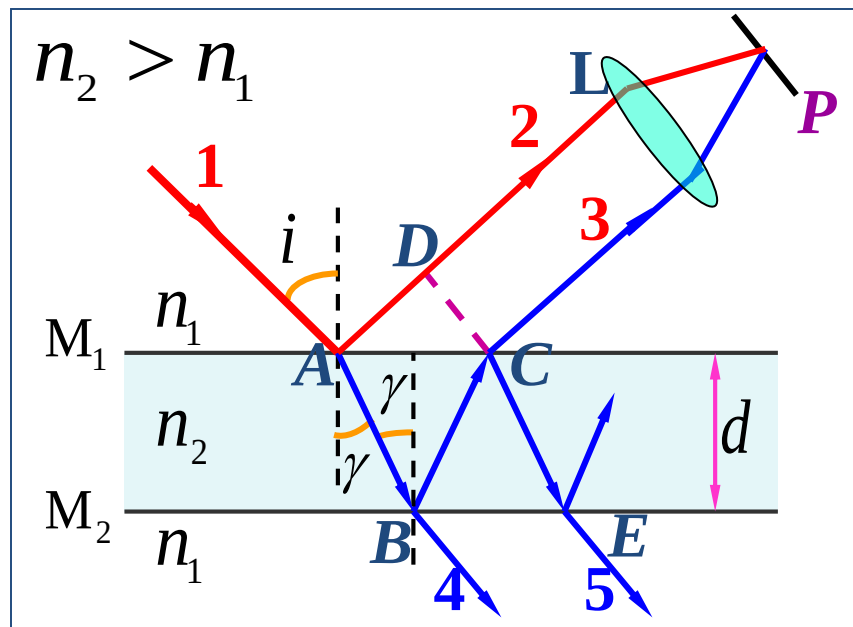




$$\Delta_{32} = \frac{2d}{\cos \gamma} n_2 (1 - \sin^2 \gamma) + \frac{\lambda}{2} = 2n_2 d \cos \gamma + \frac{\lambda}{2}$$

➤ 反射光的光程差 $\Delta_r = 2d \sqrt{n_2^2 - n_1^2 \sin^2 i} + \frac{\lambda}{2}$

$$\Delta_r = \begin{cases} k\lambda & \text{加强} \\ (k=1,2,\cdots) \\ (2k+1)\frac{\lambda}{2} & \text{减弱} \\ (k=0,1,2,\cdots) \end{cases}$$



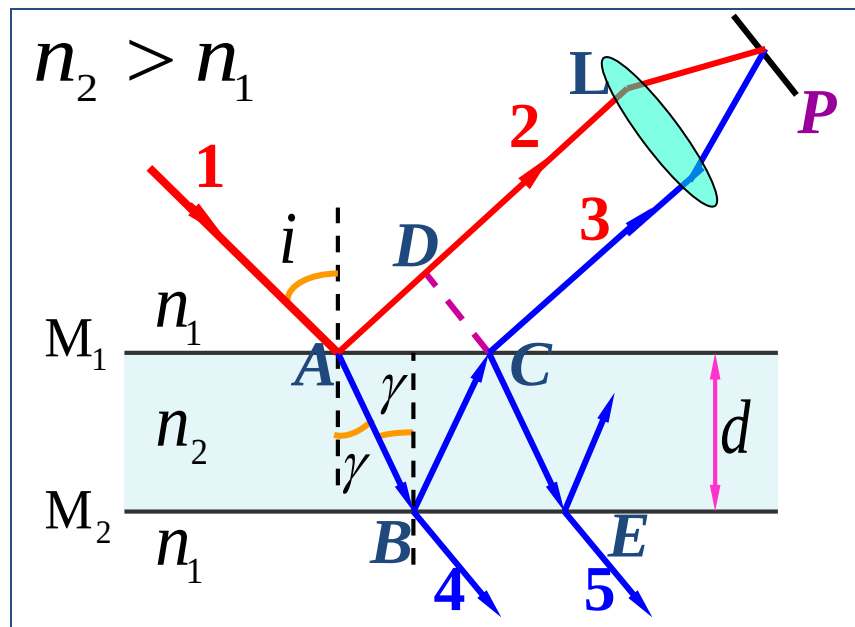


$$\Delta_r = 2d\sqrt{n_2^2 - n_1^2 \sin^2 i} + \lambda/2$$

根据具体情况而定

► 透射光的光程差 $\Delta_t = 2d\sqrt{n_2^2 - n_1^2 \sin^2 i}$

注意：透射光和反射光干涉具有互补性，符合能量守恒定律。



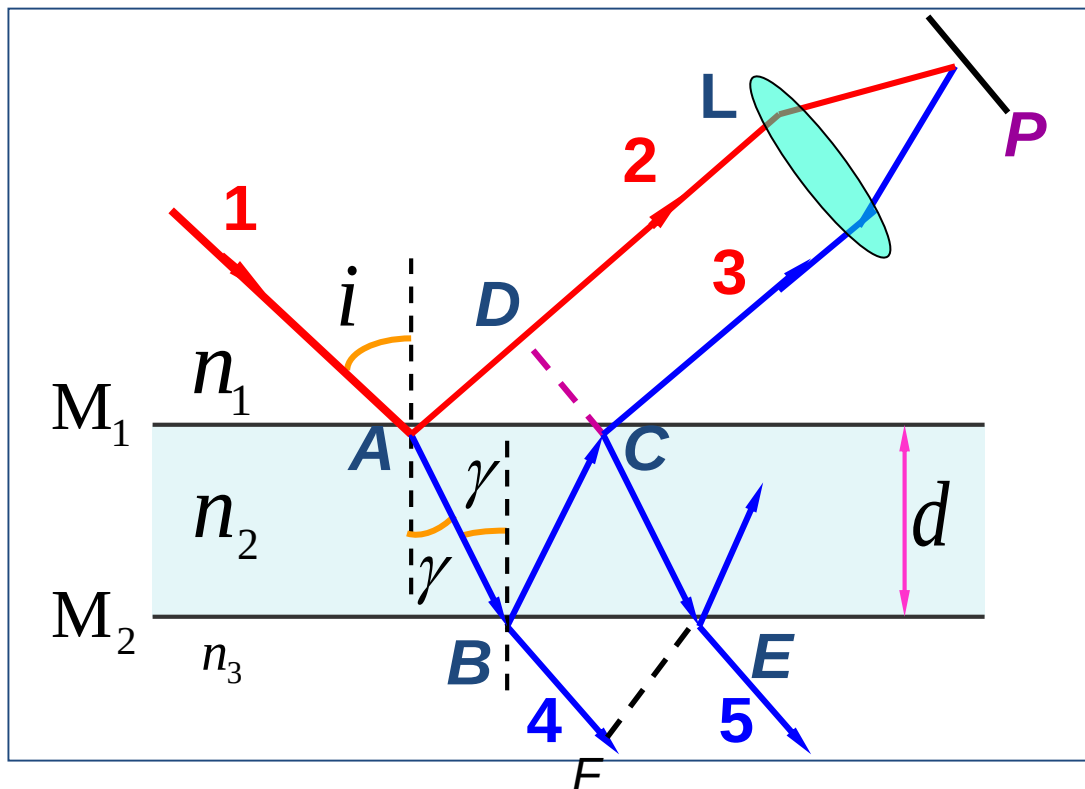


一般情况

$$n_1, n_2, n_3$$

$$CD \perp AD$$

$$EF \perp BF$$



$$n_1 \sin i = n_2 \sin \gamma = n_3 \sin i'$$



反射光的光程差

$$n_1, n_2, n_3$$

$$\Delta_r = 2d \sqrt{n_2^2 - n_1^2 \sin^2 i} + \begin{cases} \lambda/2 \\ 0 \end{cases}$$

无序排列

有序排列

透射光的光程差

$$\Delta_t = 2d \sqrt{n_2^2 - n_3^2 \sin^2 i'} + \begin{cases} 0 \\ \lambda/2 \end{cases}$$

无序排列

有序排列

$$n_1 \sin i = n_3 \sin i'$$

注意：透射光和反射光干涉具有互补性，符合能量守恒定律。



讨论

11-3 薄膜干涉

$$\Delta_r = 2d \sqrt{n_2^2 - n_1^2 \sin^2 i} + \frac{\lambda}{2} = \begin{cases} k\lambda & k = 1, 2, \dots \\ (2k+1)\frac{\lambda}{2} & k = 0, 1, 2, \dots \end{cases} \quad \begin{matrix} \text{明} \\ \text{暗} \end{matrix}$$

若 λ 、 n_1 、 n_2 一定， Δ 与 d 、 i 有关

(1) 薄膜厚度均匀(d 一定)， Δ 随入射角 i 变化

同一入射角 i 对应同一级干涉条纹

不同入射角 对应不同级次的条纹

干涉条纹为一组同心圆环

等倾干涉

(2) 入射角 i 一定(平行光入射)， Δ 随薄膜厚度 d 变化

薄膜同一厚度处对应同一级干涉条纹

薄膜不同厚度处对应不同级次干涉条纹

条纹形状与薄膜等厚线相同

等厚干涉

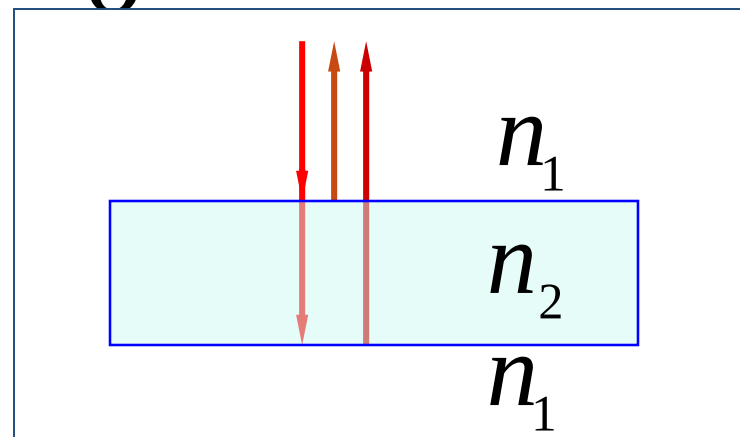


均匀膜的等倾干涉

◆ 当光线垂直入射时 $i = 0^\circ$

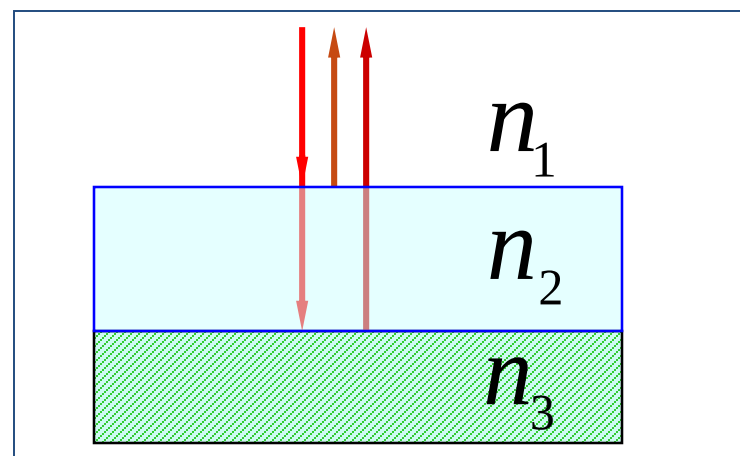
当 $n_2 > n_1$ 时

$$\Delta_r = 2dn_2 + \frac{\lambda}{2}$$



当 $n_3 > n_2 > n_1$ 时

$$\Delta_r = 2dn_2$$





例 1 一油轮漏出的油 (折射率 $n_1=1.20$) 污染了某海域, 在海水 ($n_2=1.30$) 表面形成一层薄薄的油污.

(1) 如果太阳正位于海域上空, 一直升飞机的驾驶员从机上向正下方观察, 他所正对的油层厚度为460 nm, 则他将观察到油层呈什么颜色?

(2) 如果一潜水员潜入该区域水下, 并向正上方观察, 又将看到油层呈什么颜色?



已知 $n_1=1.20$ $n_2=1.30$ $d=460 \text{ nm}$

解 (1) $\Delta_r = 2dn_1 = k\lambda$

$$\lambda = \frac{2n_1d}{k}, \quad k=1,2,\dots$$

$$k=1, \quad \lambda = 2n_1d = 1104 \text{ nm}$$

$$k=2, \quad \lambda = n_1d = 552 \text{ nm}$$

绿色

$$k=3, \quad \lambda = \frac{2}{3}n_1d = 368 \text{ nm}$$



(2) 透射光的光程差 $\Delta_t = 2dn_1 + \lambda/2$

$$k=1, \quad \lambda = \frac{2n_1d}{1-1/2} = 2208 \text{ nm}$$

紫红色 { $k=2, \quad \lambda = \frac{2n_1d}{2-1/2} = 736 \text{ nm}$ 红光

$k=3, \quad \lambda = \frac{2n_1d}{3-1/2} = 441.6 \text{ nm}$ 紫光

$$k=4, \quad \lambda = \frac{2n_1d}{4-1/2} = 315.4 \text{ nm}$$

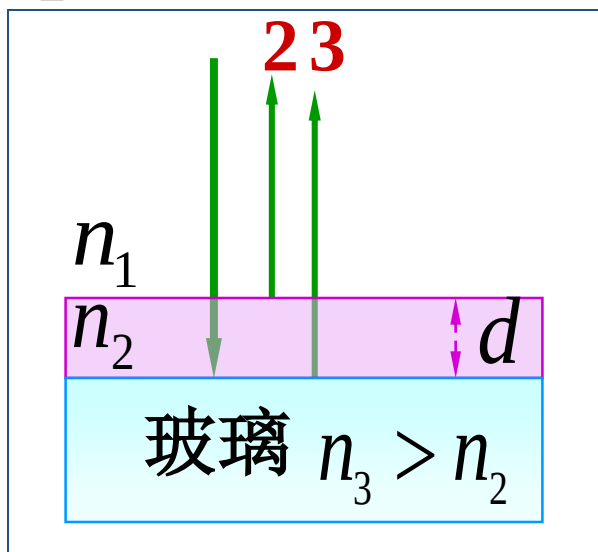


◆ 增透膜和增反膜

利用薄膜干涉可以提高光学器件的透光率。



例2 为了增加透射率，求氟化镁膜的最小厚度. 已知 空气 $n_1=1.00$ ，氟化镁 $n_2=1.38$ ， $\lambda=550\text{ nm}$



氟化镁为增透膜

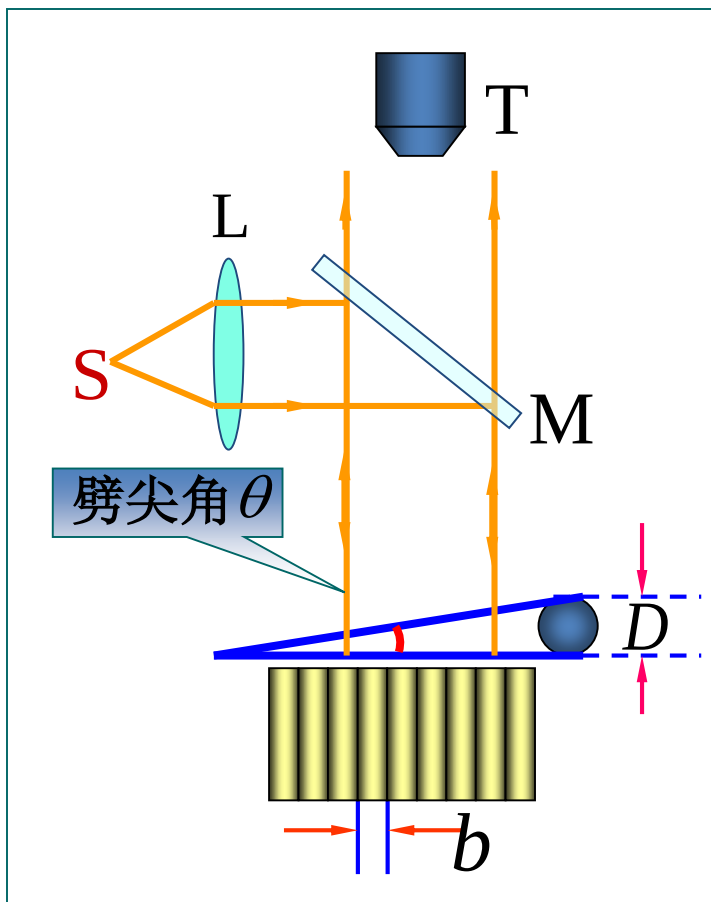
解 $\Delta_r = 2dn_2 = (2k+1)\frac{\lambda}{2}$
取 $k=0$ **减弱**

$$d = d_{\min} = \frac{\lambda}{4n_2} = 99.6\text{ nm}$$

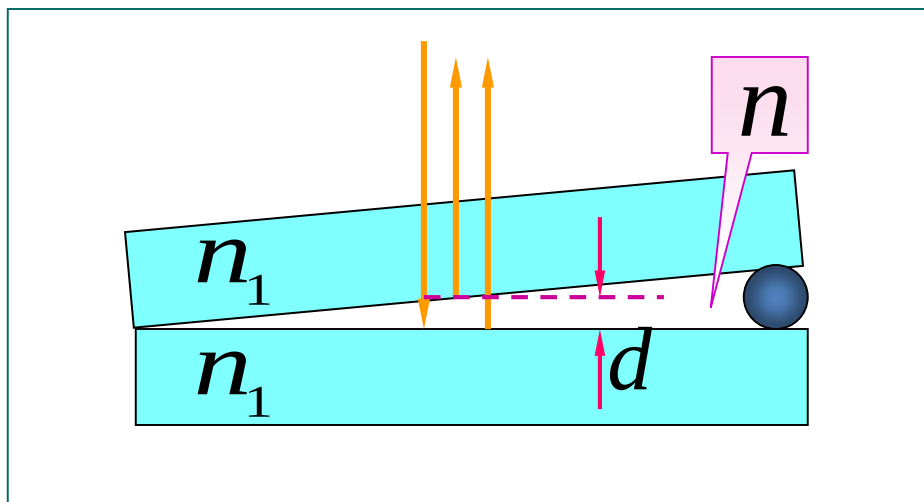
则 $\Delta_t = 2n_2d + \frac{\lambda}{2} = \lambda$ (**增强**)



一 劈尖



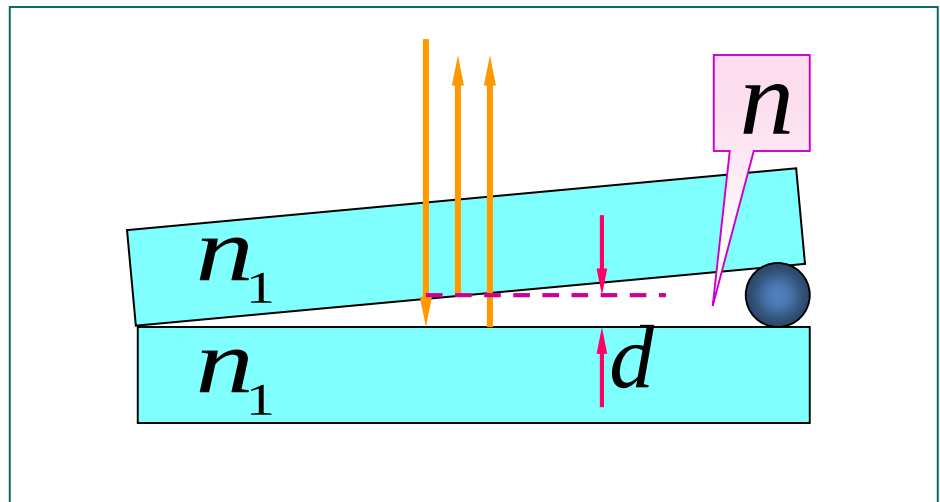
$$\Delta = 2nd + \frac{\lambda}{2}$$





$$\Delta = 2nd + \frac{\lambda}{2}$$

$$\Delta = \begin{cases} k\lambda, & k = 1, 2, \dots \quad \text{明纹} \\ (2k+1)\frac{\lambda}{2}, & k = 0, 1, \dots \quad \text{暗纹} \end{cases}$$



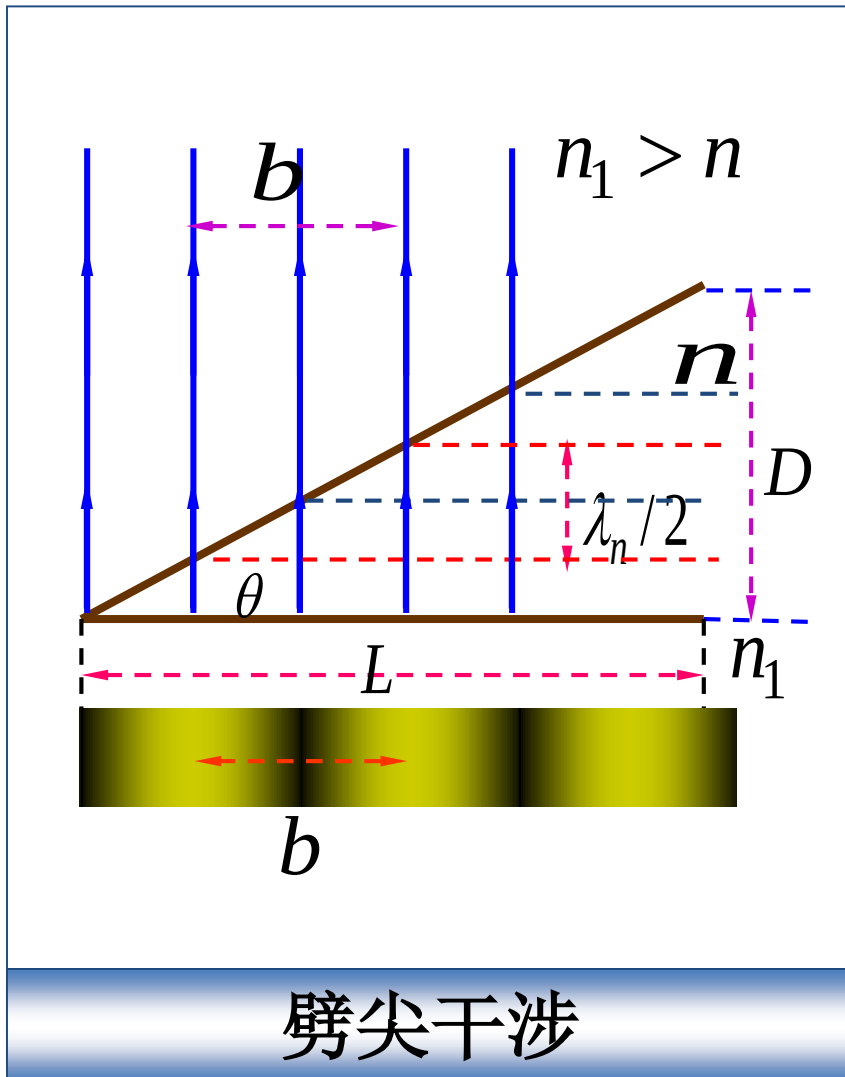


讨论

(1) 棱边处 $d = 0$

$\Delta = \frac{\lambda}{2}$ 为暗纹.

$$d = \begin{cases} (k - \frac{1}{2}) \frac{\lambda}{2n} & \text{(明纹)} \\ k\lambda/2n & \text{(暗纹)} \end{cases}$$



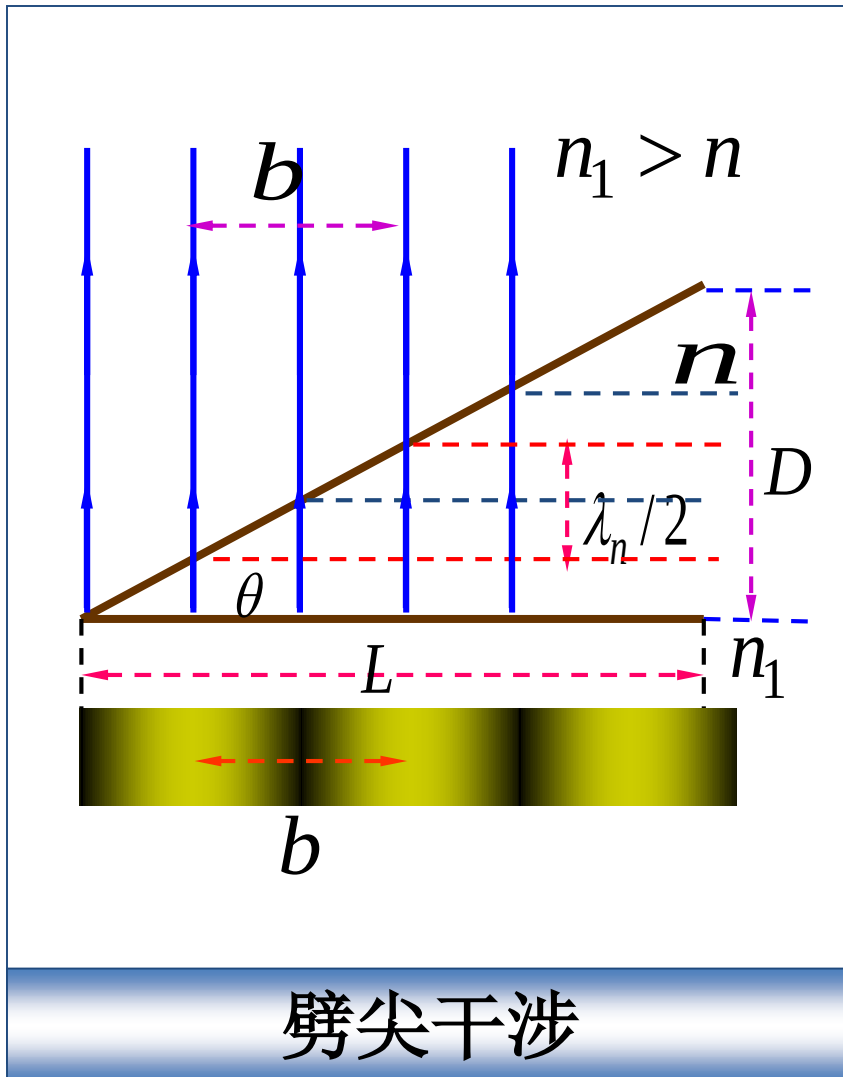


(2) 相邻明纹（暗纹） 间的厚度差

$$d_{i+1} - d_i = \frac{\lambda}{2n} = \frac{\lambda_n}{2}$$

$$\theta \approx D/L$$

$$\theta \approx \frac{\lambda_n/2}{b}$$

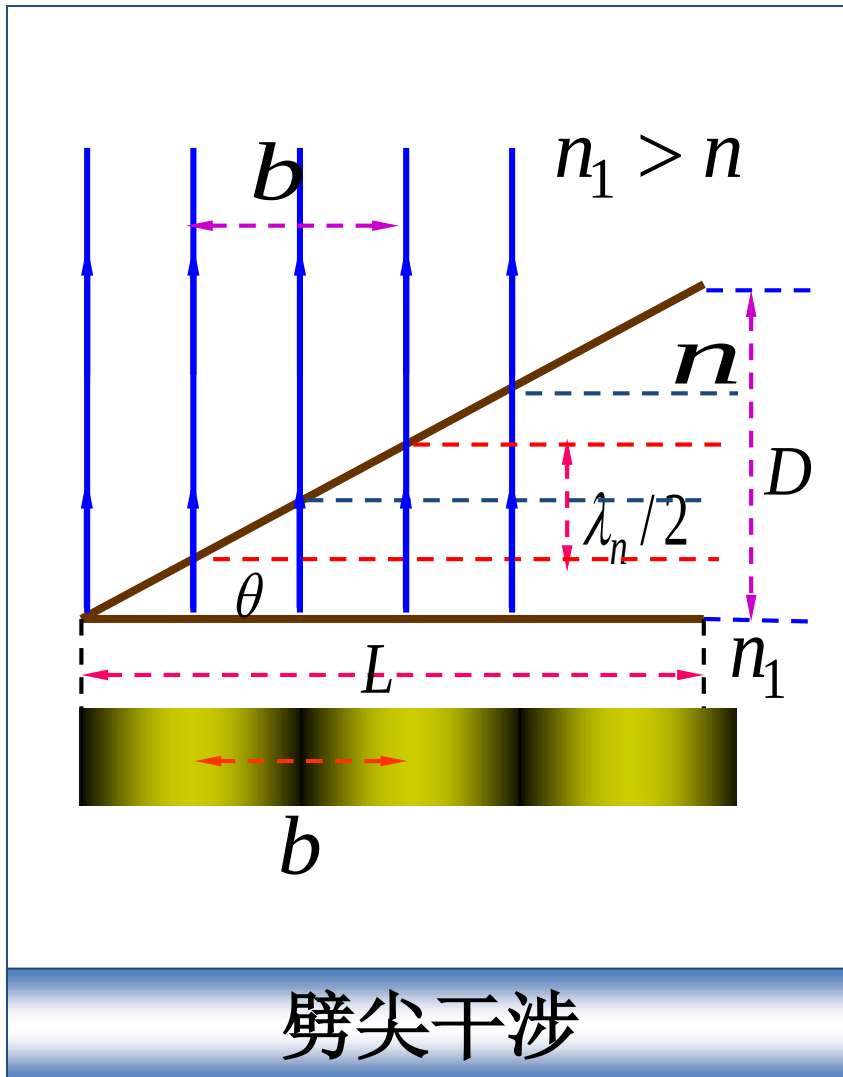




(3) 条纹间距

$$b = \frac{\lambda}{2n\theta}$$

$$D = \frac{\lambda_n}{2b} L = \frac{\lambda}{2nb} L$$





例 1 波长为680 nm的平行光照射到 $L=12\text{ cm}$ 长的两块玻璃片上，两玻璃片的一边相互接触，另一边被厚度 $D=0.048\text{ mm}$ 的纸片隔开。试问在这12 cm长度内会呈现多少条暗条纹？

解

$$2d + \frac{\lambda}{2} = (2k + 1) \frac{\lambda}{2}$$
$$k = 0, 1, 2, \dots$$



$$2d + \frac{\lambda}{2} = (2k + 1)\frac{\lambda}{2} \quad k = 0, 1, 2, \dots$$

$$2D + \frac{\lambda}{2} = (2k_m + 1)\frac{\lambda}{2}$$

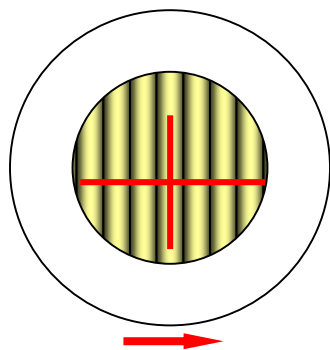
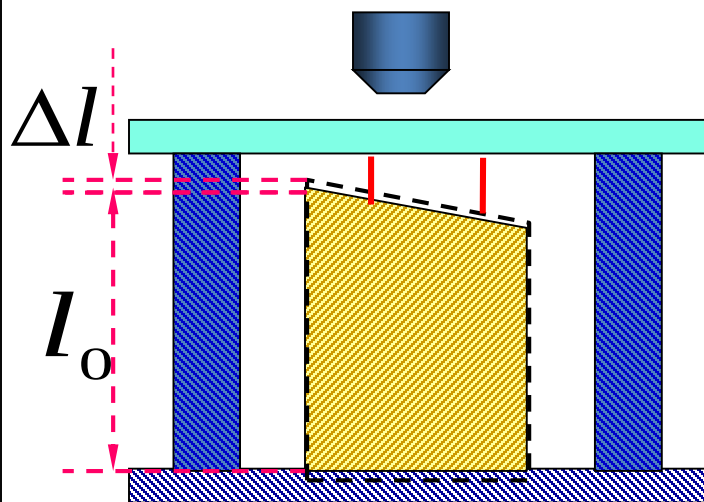
$$k_m = \frac{2D}{\lambda} = 141.2$$

共有142条暗纹



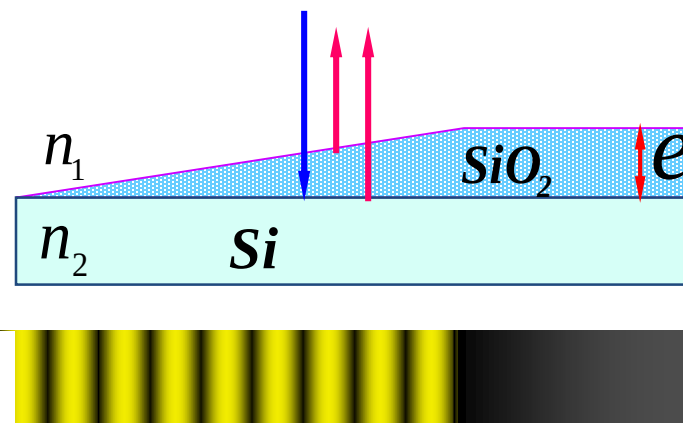
劈尖干涉的应用

(1) 干涉膨胀仪



$$\Delta l = N \frac{\lambda}{2}$$

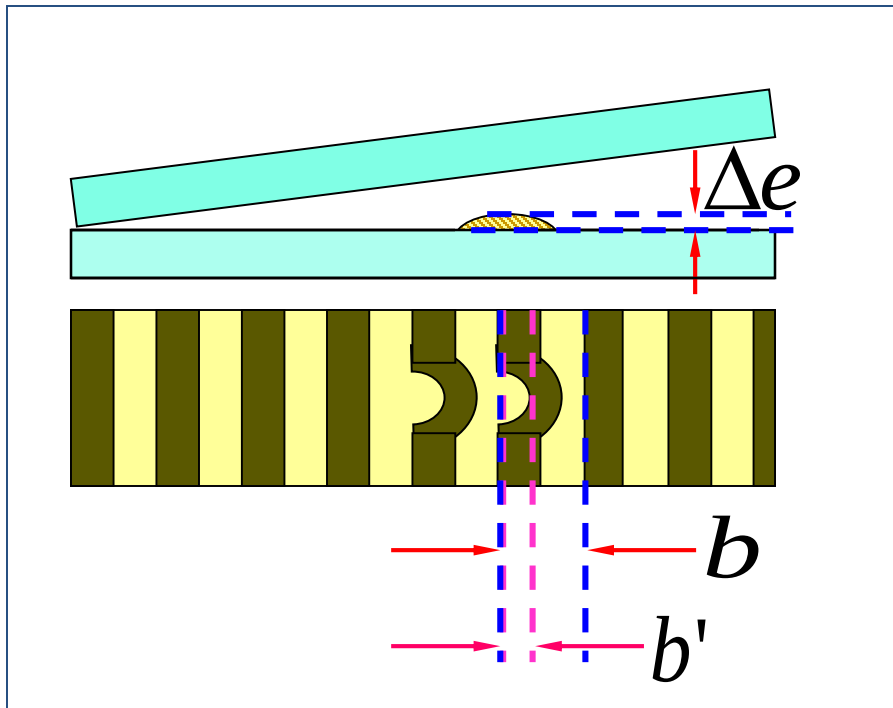
(2) 测膜厚



$$e = N \frac{\lambda}{2n_1}$$



(3) 检验光学元件表面的平整度

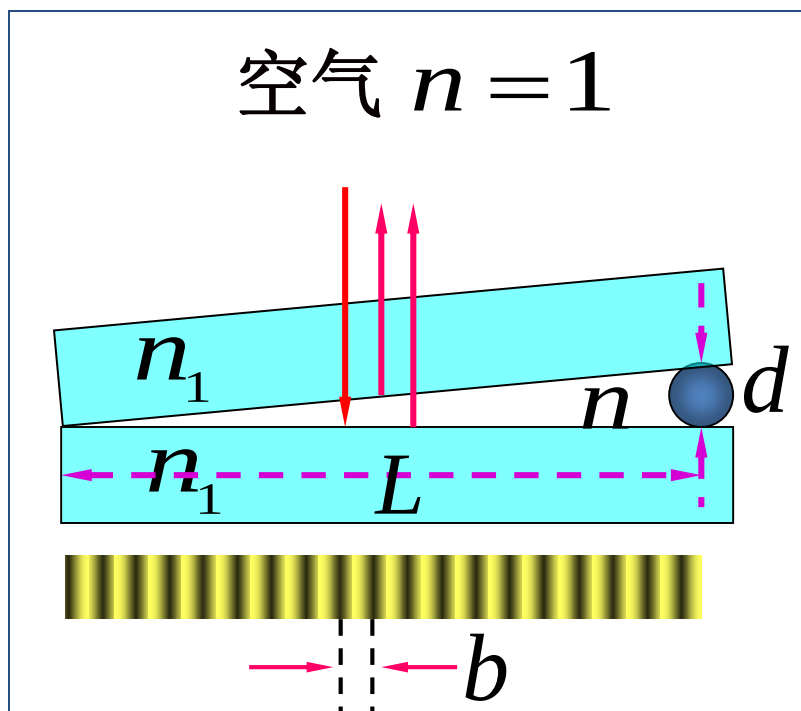


$$\Delta e = \frac{b'}{b} \frac{\lambda}{2}$$

$$\approx \frac{1}{3} \cdot \frac{\lambda}{2} = \frac{\lambda}{6}$$



(4) 测细丝的直径

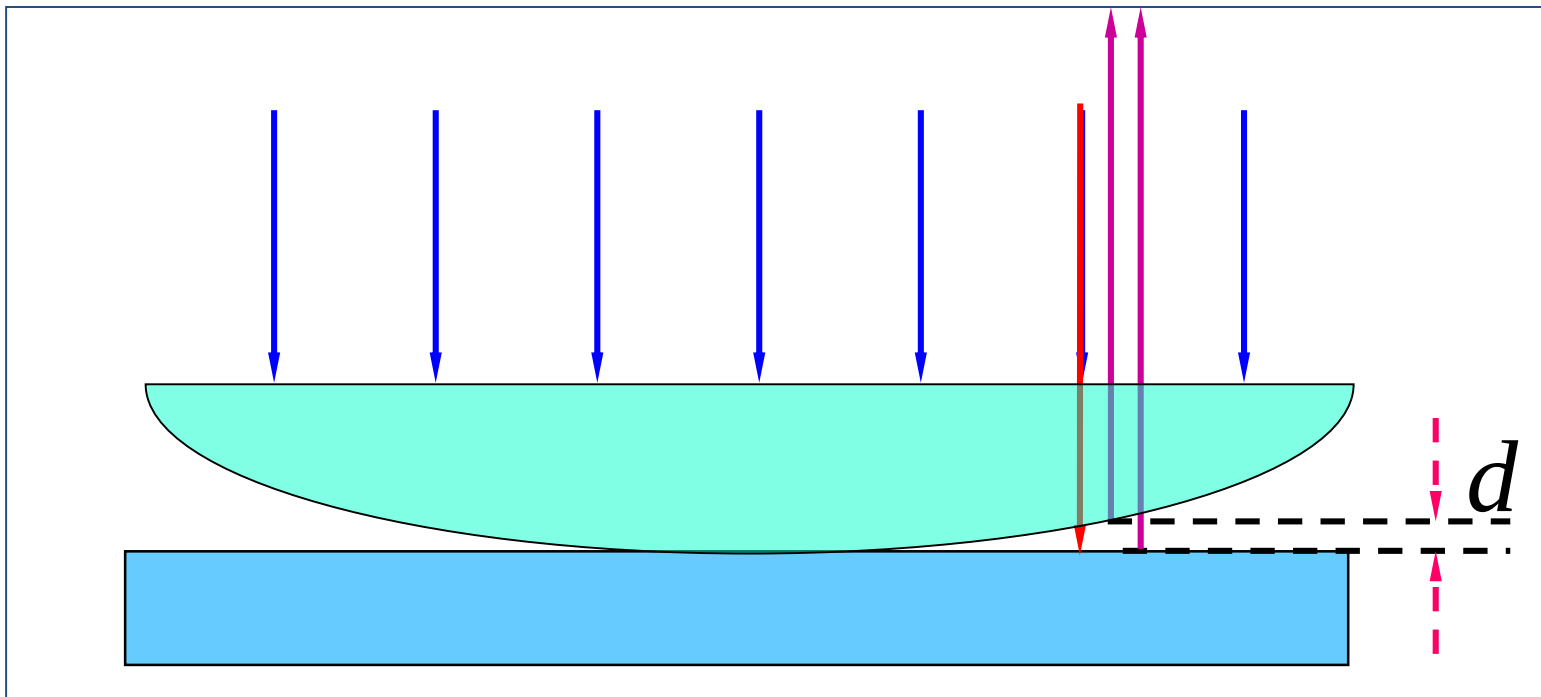


$$d = \frac{\lambda}{2n} \cdot \frac{L}{b}$$



二 牛顿环

由一块平板玻璃和一平凸透镜组成

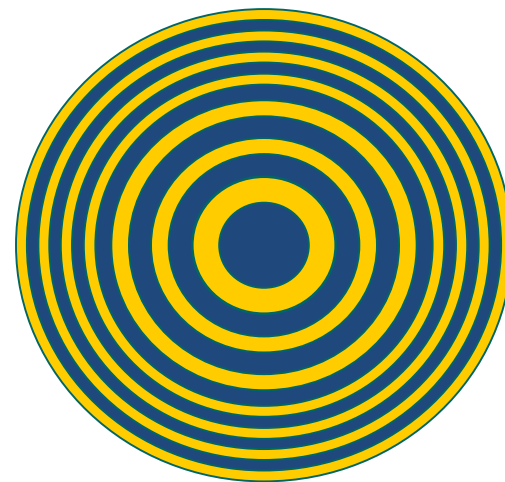
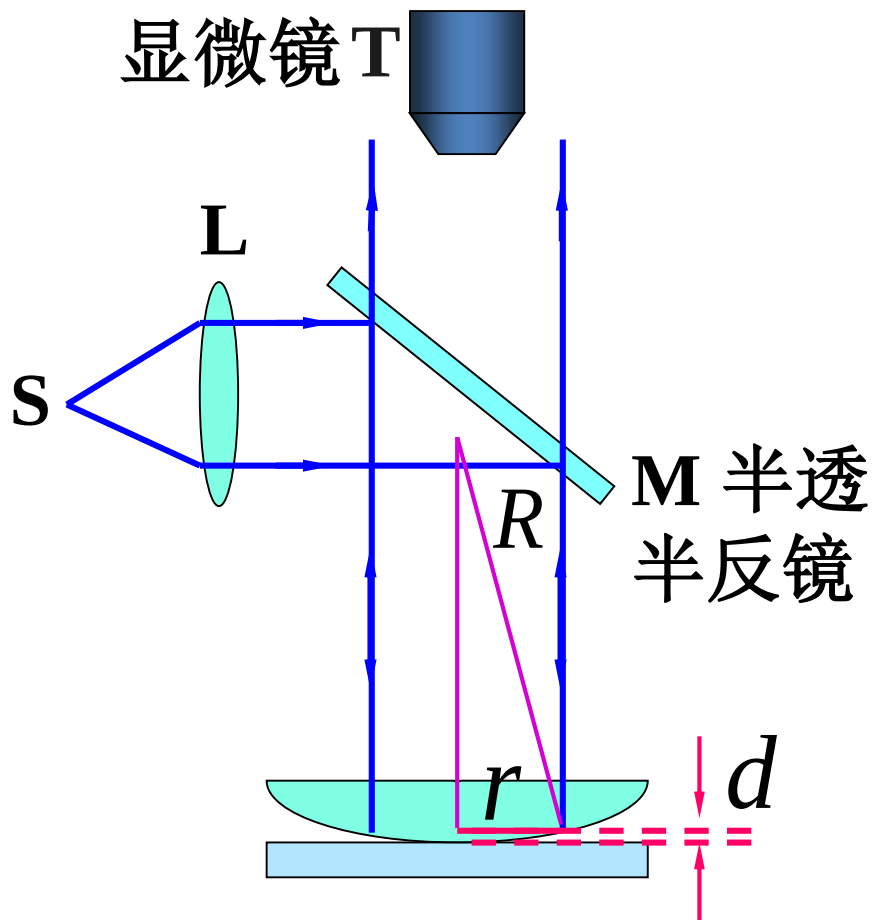


光程差

$$\Delta = 2d + \frac{\lambda}{2}$$



牛顿环实验装置



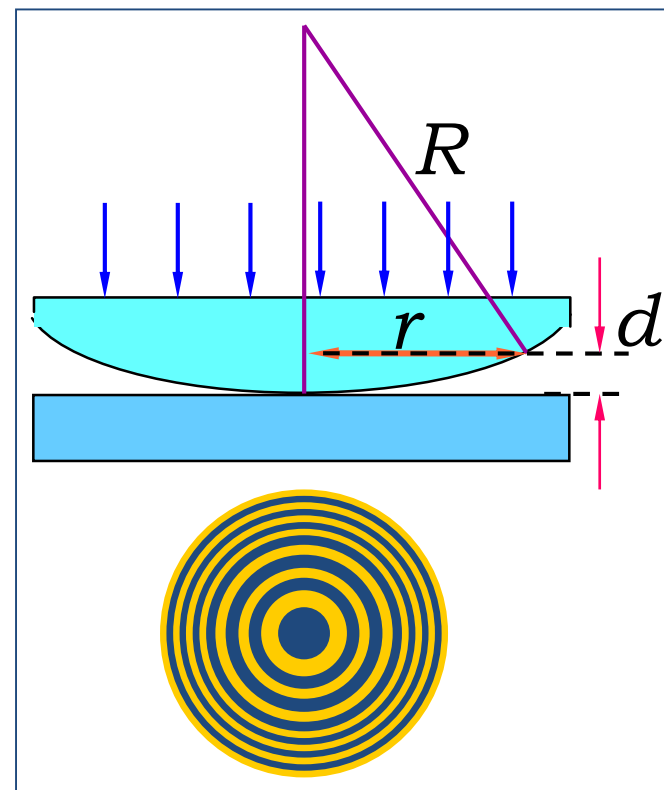
牛顿环干涉图样



光程差

$$\Delta = 2d + \frac{\lambda}{2}$$

$$\Delta = \begin{cases} k\lambda & (k = 1, 2, \dots) \quad \text{明纹} \\ (k + \frac{1}{2})\lambda & (k = 0, 1, \dots) \quad \text{暗纹} \end{cases}$$



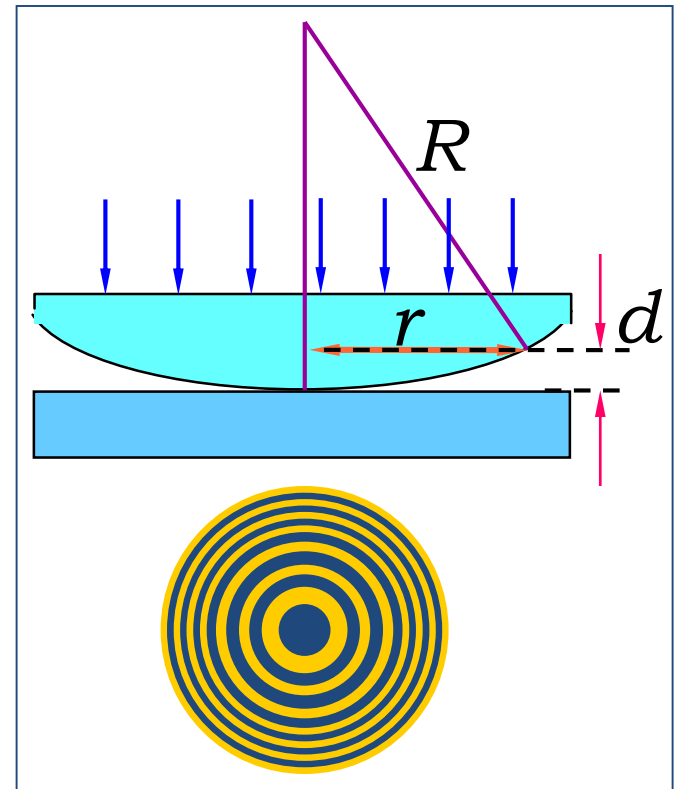


$$r^2 = R^2 - (R - d)^2 = 2dR - d^2$$

$$\because R \gg d \quad \therefore d^2 \approx 0$$

$$r = \sqrt{2dR} = \sqrt{\left(\Delta - \frac{\lambda}{2}\right)R}$$

$$\Rightarrow \begin{cases} r = \sqrt{\left(k - \frac{1}{2}\right)R\lambda} & \text{明环半径} \\ r = \sqrt{kR\lambda} & \text{暗环半径} \end{cases}$$





讨论

明环半径 $r = \sqrt{(k - \frac{1}{2})R\lambda} \quad (k = 1, 2, 3, \dots)$

暗环半径 $r = \sqrt{kR\lambda} \quad (k = 0, 1, 2, \dots)$

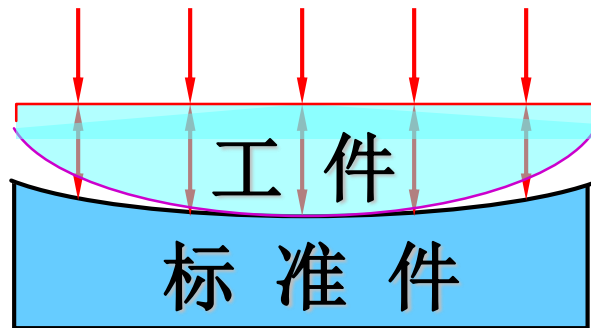
(1) 从反射光中观测，中心点是暗点还是亮点？从透射光中观测，中心点是暗点还是亮点？

(2) 属于等厚干涉，条纹间距不等，为什么？



(3) 将牛顿环置于 $n > 1$ 的液体中，条纹如何变？

(4) 应用例子：可以用来测量光波波长，用于检测透镜质量，曲率半径等。



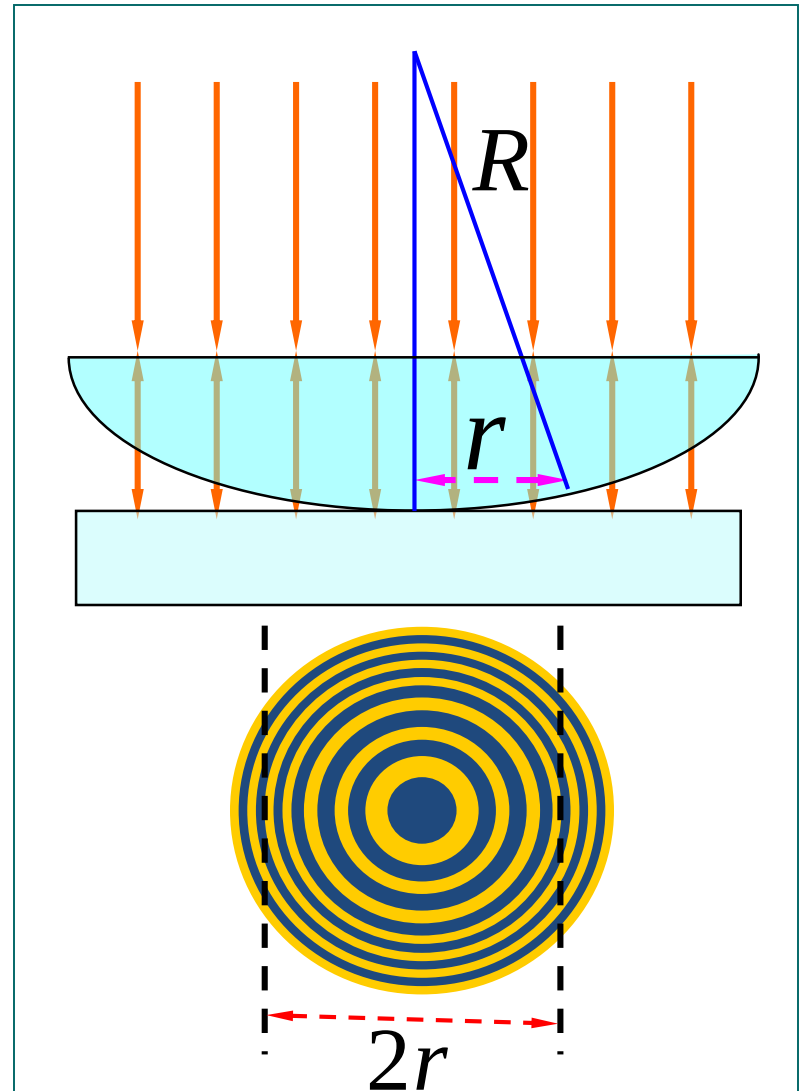


◆ 测量透镜的曲率半径

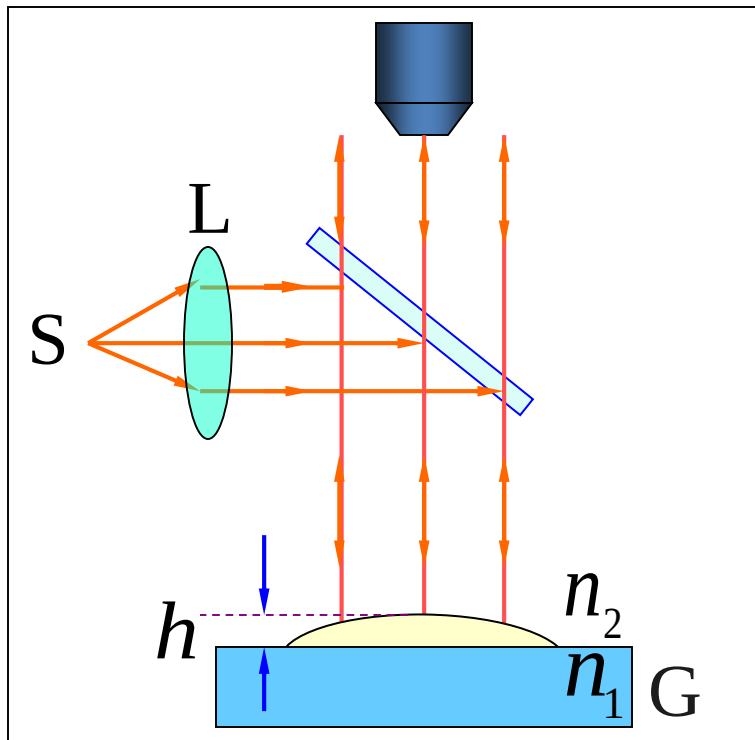
$$r_k^2 = kR\lambda$$

$$r_{k+m}^2 = (k+m)R\lambda$$

$$R = \frac{r_{k+m}^2 - r_k^2}{m\lambda}$$



例2 如图所示为测量油膜折射率的实验装置，在平面玻璃片G上放一油滴，并展开成圆形油膜，在波长 $\lambda = 600 \text{ nm}$ 的单色光垂直入射下，从反射光中可观察到油膜所形成的干涉条纹．已知玻璃的折射率为 $n_1 = 1.50$ ，油膜的折射率 $n_2 = 1.20$ ，问：当油膜中心最高点与玻璃



下，从反射光中可观察到油膜所形成的干涉条纹．已知玻璃的折射率为 $n_1 = 1.50$ ，油膜的折射率 $n_2 = 1.20$ ，问：当油膜中心最高点与玻璃

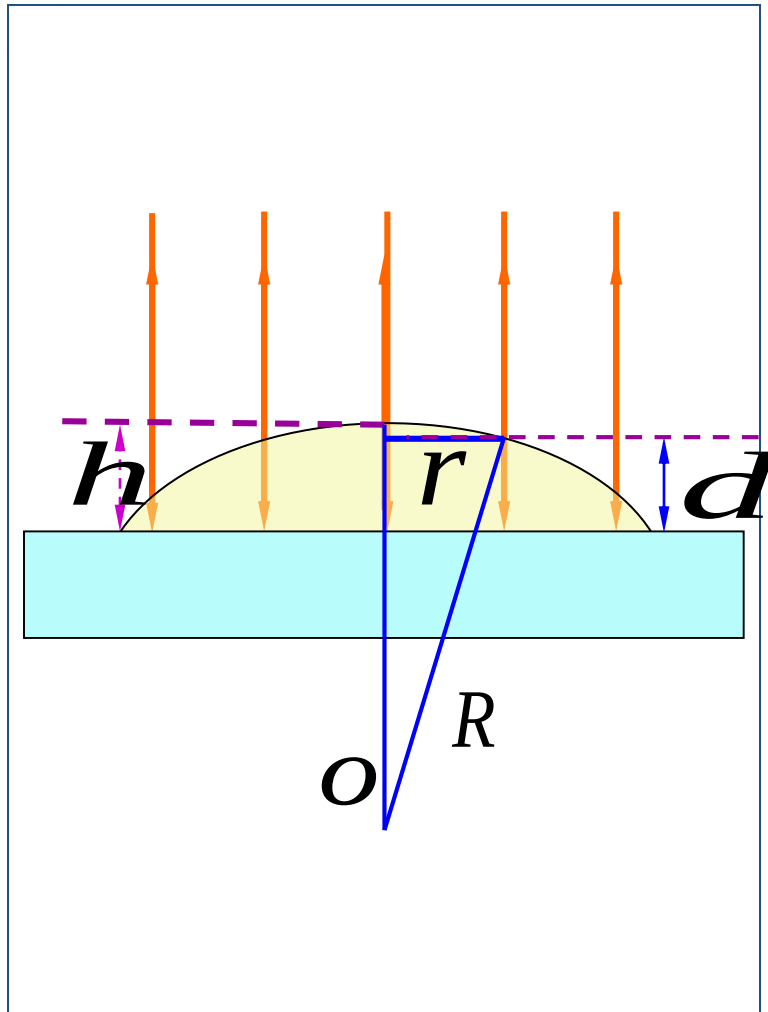

$$\Delta = 2n_2 d_k = k\lambda \quad \text{明纹}$$

$$d_k = k \frac{\lambda}{2n_2}$$

$$k = 0, 1, 2, \dots$$



11-4 劈尖 牛顿环 迈克尔孙干涉仪



油膜边缘 $k = 0, d_0 = 0$

$k = 1, d_1 = 250\text{nm}$

$k = 2, d_2 = 500\text{nm}$

$k = 3, d_3 = 750\text{nm}$

$k = 4, d_4 = 1000\text{nm}$

由于 $h = 8.0 \times 10^2 \text{nm}$ 故
可观察到**四条明纹**。



讨论

油滴展开时，条纹间距变大，条纹数减少

$$R^2 = r^2 + [R - (h - d)]^2$$

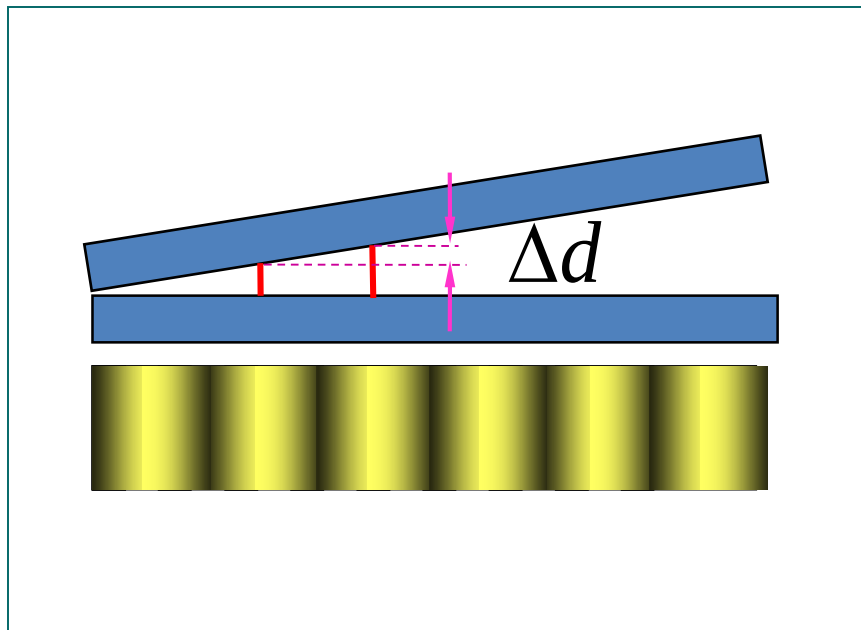
$$r^2 \approx 2R(h - d)$$

$$R \approx \frac{r^2}{2(h - d)}$$



◆ 总结

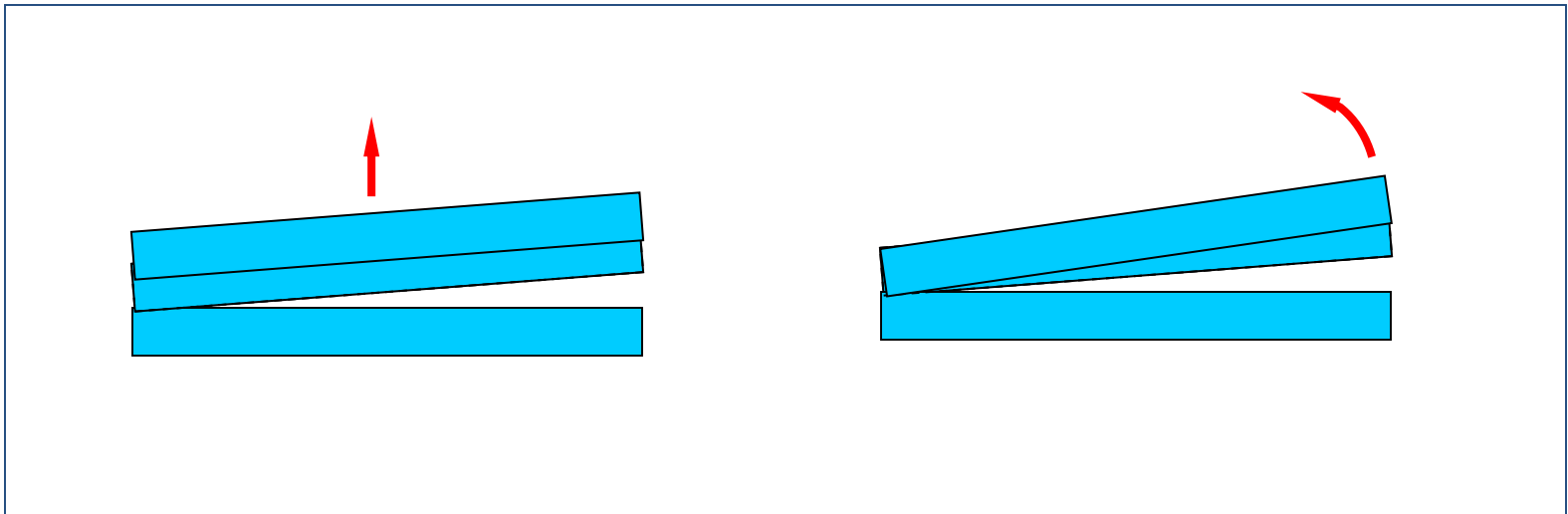
(1) 干涉条纹为光程差相同的点的轨迹，
即厚度相等的点的轨迹。



$$\Delta k = 1$$

$$\Delta d = \frac{\lambda}{2n}$$

- (2) 厚度线性增长条纹等间距，厚度非线性增长条纹不等间距。
- (3) 条纹的动态变化分析 (n, λ, θ 变化时)





(4) 半波损失需具体问题具体分析.

